

① NMAP Network Scanning Without permission:-

Analysis:-

The student's action is unethical and potentially illegal because the network scan was conducted without the company's authorization. Although Nmap is a legitimate network security tool used by cybersecurity professionals, using it on systems without permission violates ethical hacking principles.

Ethical Hacking requires prior written permission from the system owner before any security testing begins. Even if no data is stolen or systems are damaged, unauthorized scanning may be treated as an attempt to gather information for future attacks.

Relevant IT Act, 2000 provisions

Section 43 : Unauthorized access or attempts to access computer systems may attract civil liability and the compensation to.

Section 66 : If unauthorized access is performed dishonestly or fraudulently, it becomes a criminal offence punishable with imprisonment and/or fine.

Section 66B : Applies if stolen computer resources or devices are involved.

Possible Legal Consequences:-

- 1) Investigation by the cybercrime police
- 2) Civil compensation claims by the company
- 3) Criminal prosecution if malicious intent is established
- 4) Academic disciplinary action affecting the students career

Correct procedure:-

Before performing any security assessment, the student should have:

- 1) Obtained written permission from organization
- 2) Defined the Scope of testing
- 3) Signed a legal authorisation or Rules of Engagement document
- 4) Submitted a confidential vulnerability report after completion

Conclusion:-

The students action were technically educational but ethically incorrect and legally risky because authorisation wasnot obtained.

② Public Disclosure of SQL Injection Vulnerability:

Analysis:

The researcher acted unethically by immediately publishing the vulnerability and proof of concept code before the company had an opportunity to fix the issue. This exposed customers to significant cyber risks.

Responsible Vulnerability Disclosure:

Responsible disclosure is a process where:

- 1) The researcher privately informs the organisation
- 2) Sufficient time is provided to fix the vulnerability
- 3) Public disclosure occurs only after a patch is available or after agreed disclosure period

Risks of premature Disclosure:

- 1) Attackers can exploit the vulnerability immediately
- 2) Customer data may be stolen
- 3) Increased risk of identity theft and fraud

Legal perspective:

If the researcher accessed systems or customer data without authorization, the following may apply:

- 1) Section 43 - Unauthorized access
- 2) Section 66 - Computer-related offences

Recommended Steps:-

- 1) Verify the vulnerability carefully
- 2) Report it privately to the organisation's security team
- 3) Allow adequate time of remediation
- 4) Coordinate public disclosure after vulnerability is fixed

Conclusion:-

Responsible disclosure protects both organisations and users while encouraging ethical cybersecurity research

③ Phishing Attack on Financial Organisation:-

Type of Cyberattack

This incident is phishing attack, specifically a credential phishing attack

The attacker impersonated the company's IT support department and tricked the employee into revealing login credentials

Legal Implications under the IT Act:-

The attacker may be prosecuted under:

- 1) Section 43 - Unauthorised access and damage
- 2) Section 66 - Computer-related offences
- 3) Section 66C - Identity theft credentials

Section 66D - cheating by personation using computer resources

Organization's Responsibilities:-

- Conduct employee security awareness training
- Implement Multi Factor Authentication
- Deploy email filtering and anti-phishing solution
- Monitor suspicious login activities
- Maintain incident response procedures

Employee's Responsibilities:-

- Verify suspicious emails
- Avoid clicking unknown links
- Report phishing attempts immediately
- Follow company cybersecurity policies
- Use strong and unique passwords

Prevention Measures:-

Technical Measures:-

- Multi Factor Authentication (MFA)
- Email authentication
- Endpoint Security
- Anti-phishing software

Organizational Measures:-

- Regular cybersecurity awareness programs
- Security Audits
- Password Management Policies
- Incident Response planning

Conclusion:-

Human error is often the weakest security link. Combining technical controls with user awareness significantly reduces phishing risks

④ Exploiting a Banking Vulnerability Without Permission:-

Analysis:-

Although the student's intention was to improve security, exploiting the vulnerability and retrieving customer information without authorization is both unethical and potentially illegal

Applicable IT Act provisions:-

- 1) Section 43 - Unauthorized access and copying of data
- 2) Section 66 - Computer related offences
- 3) Section 72 - Breach of confidentiality if customer information is accessed

Ethical Hacking vs Unauthorized Hacking :-

Ethical Hacking	Unauthorized Hacking
Permission obtained	No permission
Legal activity	Illegal activity
Conducted within agreed scope	Scope not defined
Improve security	violates security policies
Responsible disclosure	Unauthorized exploitation

Proper Reporting Process :-

- 1) Prepare detailed technical Report
- 2) Donot exploit the vulnerability beyond verification
- 3) Request secure communication for disclosure
- 4) Allow sufficient time for remediation before any of the public disclosure

Conclusion :-

Good intentions donot make unauthorized access legal. Ethical Hackers must obtain authorization before testing